Predicting the Duration of Eye-fixations on Natural Images Using Deep Neural Networks

Ira Ceka[1], Xavier Boix[2]

[1] University of Massachusetts, Boston

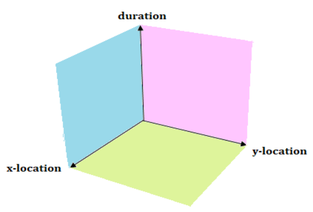
[2] Center for Brains, Minds, and Machines, CBMM, MIT

Introduction

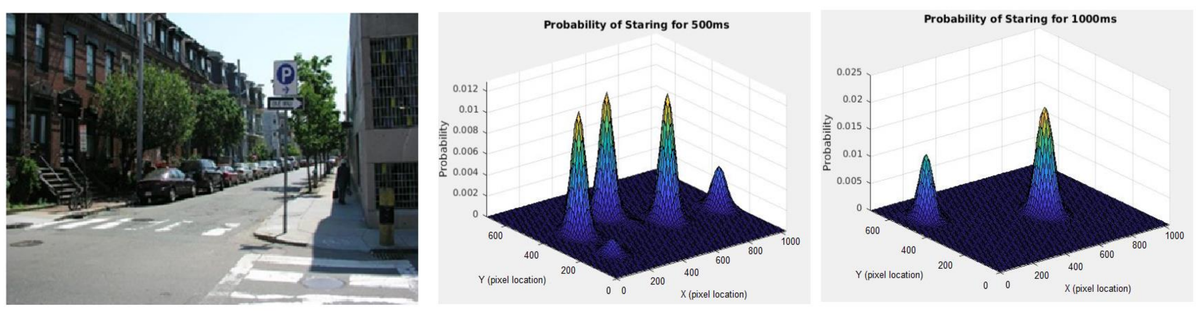
- **Motivation**: Predicting duration → understanding human visual attention
- **Data**: MIT1003 dataset [1]
  - 1003 natural images of indoor/outdoor scenes

Figure 1: Sample images from the MIT1003 dataset.

- **Representing Duration**: encompassing information about location and duration into one 3D structure



3D plot —Gaussian Filter→ **Probability Distribution**



(1) (2) (3)

Figure 2: **(1)** A sample image from the MIT1003 dataset and two slices [**(2)**, **(3)**] from its associated 3D map. **(2)** represents the probability distribution of fixating on a certain location for 500 milliseconds, **(3)** for 1 second.

Methods

- **Network Architecture**: adapting a VGG16[2] model pretrained on ImageNet for object recognition
  - **Input**: **Image**
  - **Label**: **3D structure**

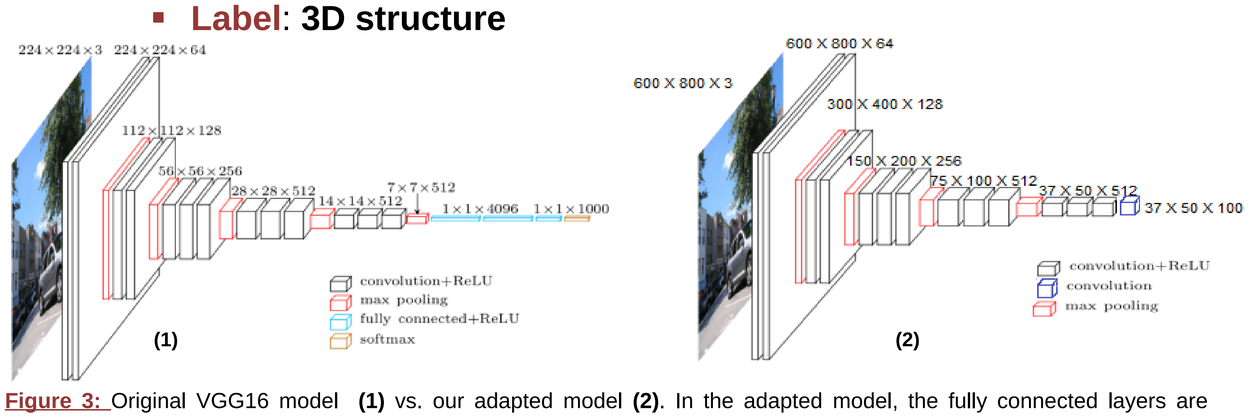


Figure 3: Original VGG16 model **(1)** vs. our adapted model **(2)**. In the adapted model, the fully connected layers are removed and replaced with a 1X1 convolutional layer. Parameters are loaded from a VGG16 model pre-trained for object recognition, after which only the last layer is trained.

Methods

- **Training**: adapting a VGG16[2] model pretrained on ImageNet for object recognition,
  - 703-image training set
  - **Learning Parameters**:
    - 10^{-5} learning rate
    - 0.9 momentum,
    - 10^{-13} weight decay
    - batch size of 1
  - **Network Parameters**:
    - 1 convolutional layer
    - 1x1 kernel size
    - No RELU
    - Filter size of 100

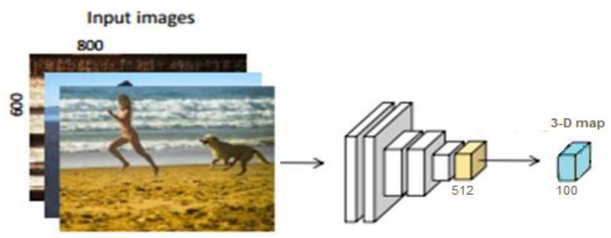
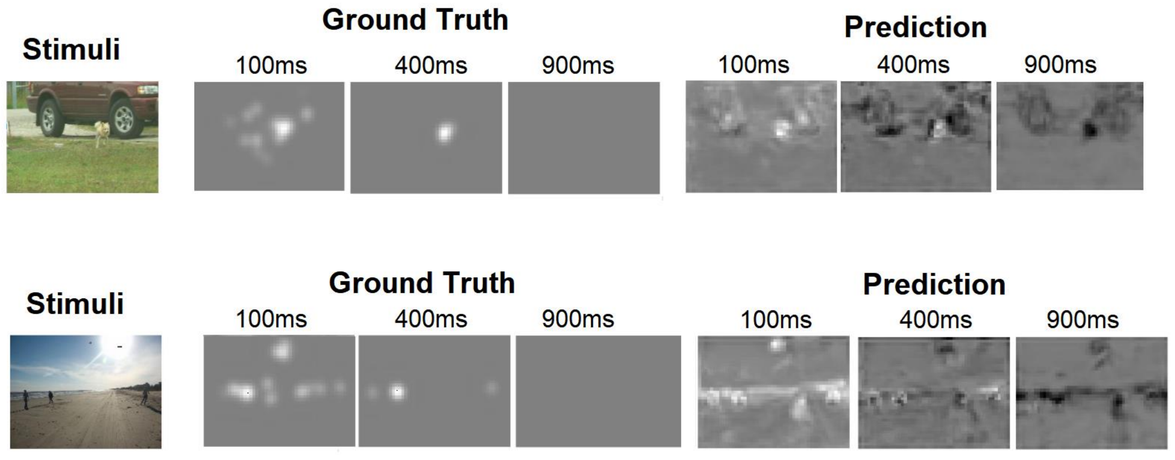


Figure 4: Only the last convolutional layer is used in training. It has a kernel size of 1x1 and a filter size of 100, returning the 100 slices of the 3D structure representing a duration instance.

Results



Conclusion & Future Work

The comparative results against ground-truth and prediction outputs above suggest that deep neural networks pretrained for **object-recognition** are able to effectively learn features necessary for predicting duration. As these are pilot experiments, however, we can only say our approach looks promising. We are confident, though, that there seem to be patterns in human fixation duration given certain contexts (ex. affinity for female faces over male, long gaze at sign and texts or outstanding objects). This could give interesting insights with respect to human visual attention.

References

[1] T. Judd, K. Ehinger, F. Durand, and A. Torralba. Learning to predict where humans look. In ICCV, 2009

[2] K. Simonyan and A. Zisserman. Very deep convolutional networks for large-scale image recognition. In ICLR, 2015.

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